POWER BI • PORTFOLIO PROJECT

OLYMPIC INSIGHTS

USING DATA ANALYTICS

A six-page navigational Power BI dashboard built for the International Olympic Committee



57	231	155.87K	20.65K
EDITIONS	COUNTRIES	ATHLETES	MEDALS

Presented by **Yatharth Aphale**

CITIUS • ALTIUS • FORTIUS

AGENDA

The Journey

01 The Brief

Case study, problem statement and the data

02 The Build

Data model, measures and approach

03 The Dashboards

Six pages, each followed by its insights

04 The Verdict

Key findings and recommendations

01

The Brief

The case study, the question the IOC is asking, and the dataset we have to answer it with.

THE CASE STUDY

Business Case & Problem Statement

PROBLEM STATEMENT

The International Olympic Committee (IOC) wants advanced insights into athlete performance, country-wise medal trends and sport-specific dominance across Olympic Games. Using historical Olympic data, build a six-page Power BI navigational dashboard with navigation buttons to track achievements, evaluate performance KPIs and forecast future outcomes for strategic planning and athlete development.

Who asks?

IOC strategy and athlete-development teams

What is needed?

Medal trends by country, sport and athlete; demographics; the host-nation effect

What is delivered?

One Power BI report with six pages linked by navigation buttons

What is success?

Clear KPIs, slicers and insights that feed planning decisions

1 Global Medal Overview | 2 Country Insights | 3 Athlete Spotlight | 4 Sport & Event Analysis | 5 Host Country Analysis | 6 Gender & Demographics

THE DATASET

Data Brief

20.65K	155.87K	231	112	964	57
Total medals	Athletes	Countries	Sports	Events	Editions

What it covers

Historical Olympic Games from 1896 onward, Summer and Winter editions. Records span athletes, events, results, host editions and country medal tallies. The report timeline slicer runs 1896 to 2032.

Six source tables

Athlete_Bio, Athlete_Event_Results, Results, Games, Games_Medal_Tally and Country. Two fact tables (athlete results and medal tally) are described by four dimension-style tables.

Prepared fields

age_at_edition and age_band (Under 18, 18–24, 25–30, 31–40, Above 40, Unknown), Key_event, Is_Host_Country and host_status. Medals are counted per athlete award, so team medals count for every member (44.69K athlete-level vs 20.65K headline medals).

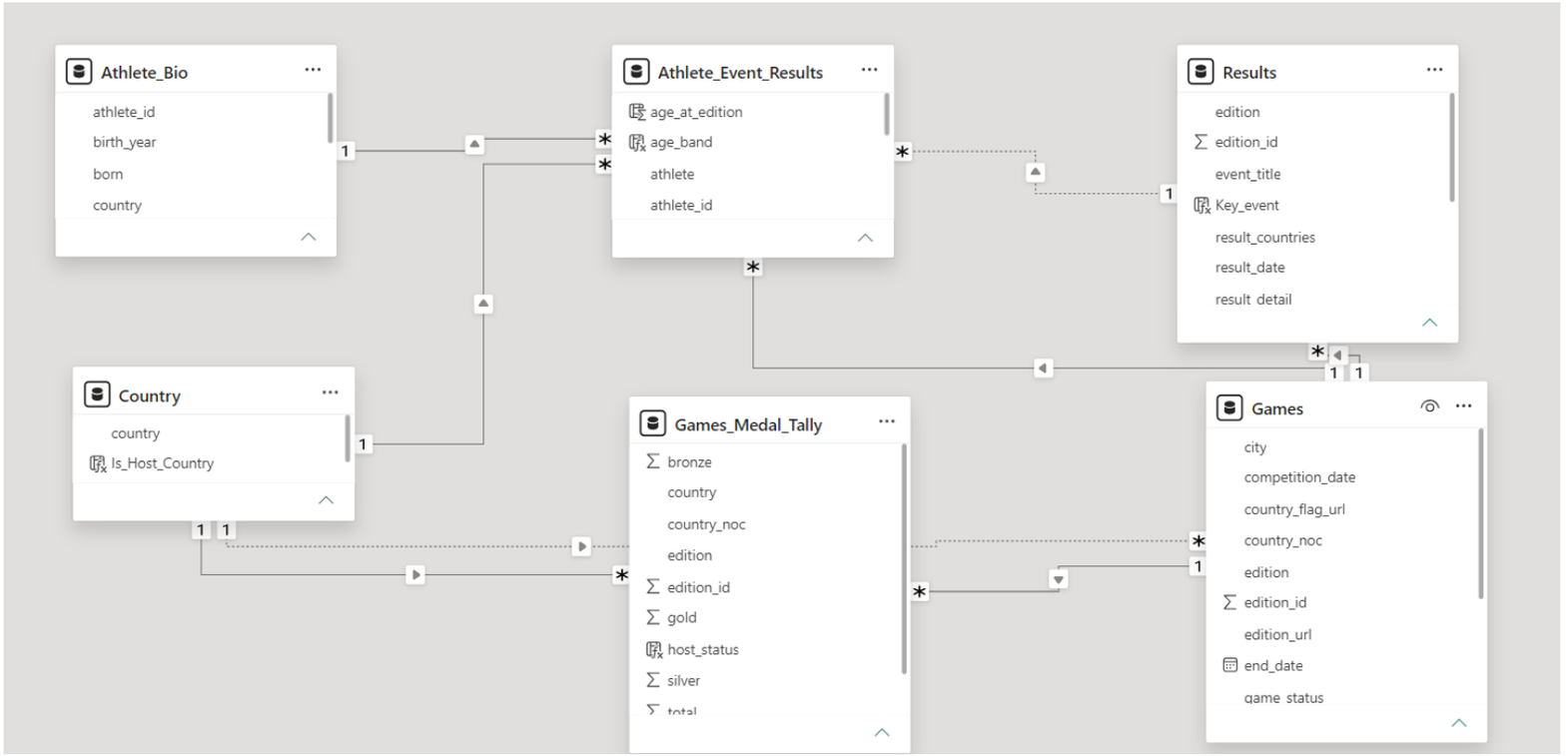
02

The Build

How the tables connect, how every KPI is defined, and the path from case study to finished report.

ARCHITECTURE

Data Model



Power BI Model view: 6 tables, solid lines = active relationships, dotted = inactive

Two fact tables

Athlete_Event_Results (participation) and Games_Medal_Tally (medals).

Shared dimensions

Country, Games and Athlete_Bio filter both facts through one-to-many links.

Inactive links

Dotted relationships stay off to avoid ambiguous filter paths.

Galaxy schema two facts sharing dimensions

age_band **Key_event** **Is_Host_Country** **host_status**

Calculated columns

UNDER THE HOOD

Tables, Roles & Keys

TABLE	TYPE	ROLE	KEY FIELDS
Athlete_Bio	Dimension	One row per athlete	*athlete_id, birth_year, born, country*
Athlete_Event_Results	Fact	Athlete participation by event and edition	*athlete_id, athlete, age_at_edition, age_band*
Results	Dimension	Event results for each edition	*edition_id, event_title, Key_event, result_date, result_detail*
Games	Dimension	Edition details: host city, dates, status	*edition_id, city, country_noc, competition_date, end_date, game_status*
Games_Medal_Tally	Fact	Gold, silver, bronze and total per country and edition	*country, country_noc, edition_id, gold, silver, bronze, total, host_status*
Country	Dimension	Country list with host flag	*country, Is_Host_Country*

Only fields visible in the Model view are listed; each table holds additional columns.

HOW THE NUMBERS ARE DEFINED

Core KPIs & Measures

Total Medals

20.65K

Count of medals awarded

Total Athletes

155.87K

Distinct athletes represented

Medal Efficiency

13.25

Total Medals ÷ Total Athletes × 100

Medal Winning %

20.26%

Medal-winning athletes ÷ Total Athletes

Avg Age of Participants

26.05

Mean athlete age at the Games

Host Medal Share

≈11%

Host-edition medals ÷ all medals

METHODOLOGY

Approach: From Case Study to Dashboard

1

Understand

Read the IOC brief and list the question each of the six pages must answer.

2

Prepare

Clean athlete, results and medal data; derive age_band, Key_event and host flags.

3

Model

Link two fact tables to shared dimensions with one-to-many relationships.

4

Measure

Write DAX for medals, athletes, efficiency, winning % and host share.

5

Visualise

Design six themed pages with KPI cards, charts and slicers.

6

Navigate

Add navigation buttons; read each page for findings and actions.

Result: one navigable report; every page opens with KPI cards, then charts, and shares the Season and Timeline slicers.

03

The Dashboards

Six pages. Each dashboard gets a full slide, followed by the insights it reveals.

SIX PAGES, ONE NAVIGATION BAR

Dashboard Architecture

1

Global Medal Overview

Medals, nations and events at a glance

2

Country Insights

Efficiency, best sports and yearly medals per nation

3

Athlete Spotlight

Top performers, longevity and medal-winner profile

4

Sport & Event Analysis

Where medals concentrate; country dominance

5

Host Country Analysis

Does hosting bring a medal advantage?

6

Gender & Demographics

Participation balance and athlete age

Each page carries the same navigation bar, Season slicer and KPI row, so a viewer can jump between views without losing context.

Global Medal Overview

A century-plus of the Olympic Games — medals, nations, and events at a glance

Citius Altius Fortius
FASTER | HIGHER | STRONGER

Total Medals
20.65K

Total Countries
231

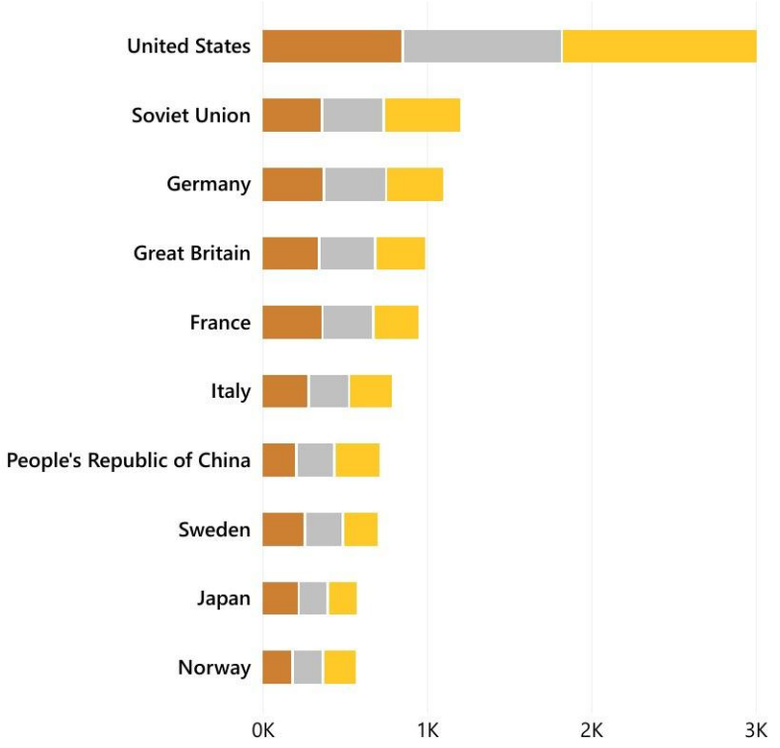
Total Events Held
7K

Total Edition Held
57

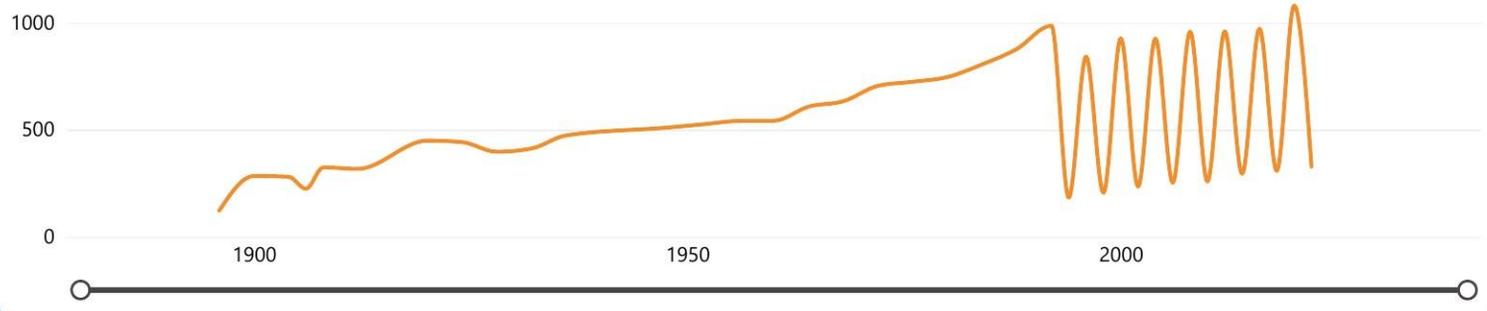
Timeline
1896 — 2032

Seasons
- Summer
- Winter

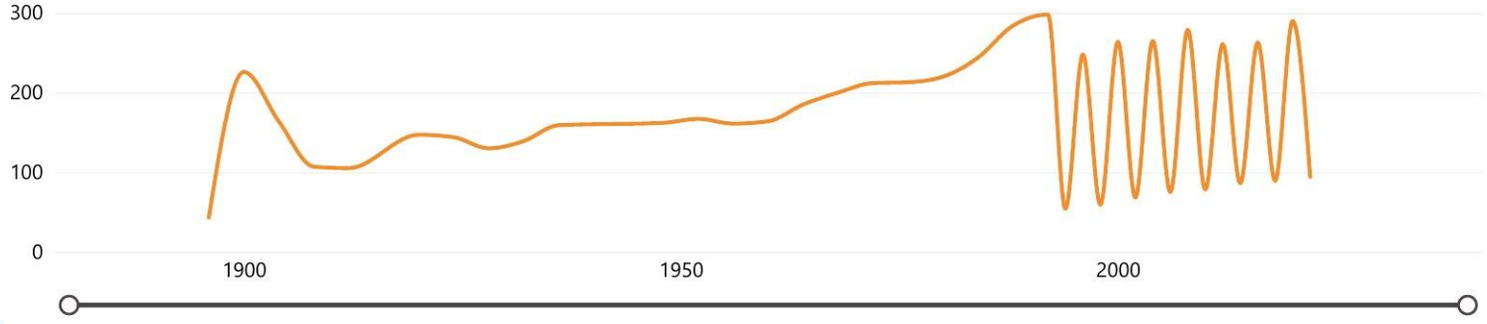
Top 10 Countries by Total Medals



Medal Trends Over the Years



Year-wise Total Events Held



Country Insights | Athlete Spotlight | Sport & Event Analysis | Host Country Analysis | Gender & Demographics

Global Medal Overview: Insights

1 A century of growth, then a Summer/Winter pulse

Medals per edition climbed from a few hundred in the early 1900s to about 1,000 by the early 1990s. Since then the line zig-zags: Summer Games reach ~1,000+ medals, Winter Games only ~200–300.

2 One nation towers over the rest

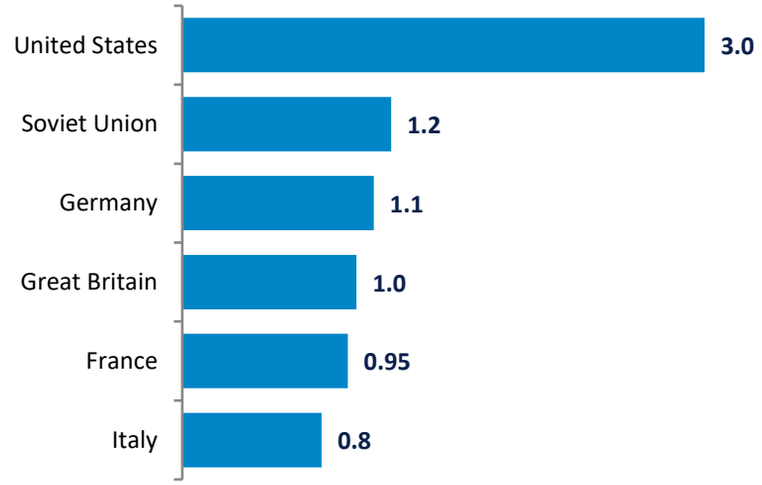
The United States leads with roughly 3K medals, about 2.5x the Soviet Union (~1.2K). Germany, Great Britain, France and Italy form a second tier near 0.8K–1.1K.

3 Programme size drives medal volume

Events per edition track the medal curve closely, peaking near 300 in Summer editions. More events means more podiums, so growth is a scheduling decision as much as a performance trend.

20.65K medals awarded across 57 editions

Top 6 countries by total medals (K, approx. from dashboard)



WHY IT MATTERS Sets the benchmark for every later page: scale, leaders and the Summer/Winter split.

Country Insights

Medal efficiency, sport strengths, and year-over-year performance by nation

Citius Altius Fortius
FASTER | HIGHER | STRONGER

Total Athlete Represented
155.87K

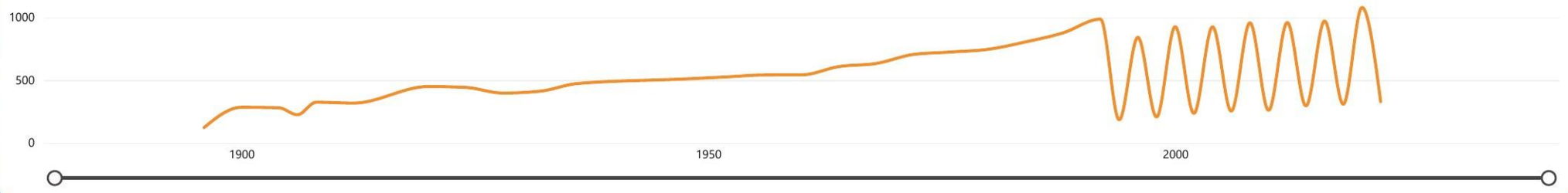
Total Medals
20.65K

Medal Efficiency
13.25

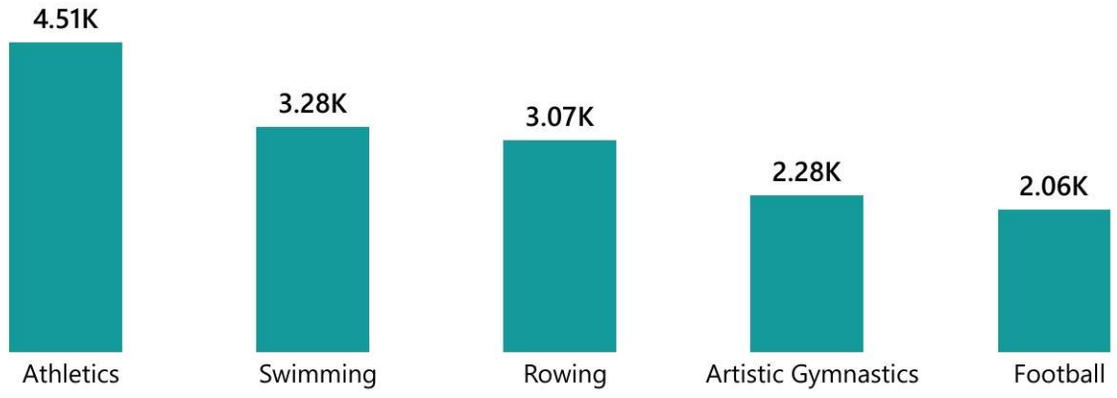
Seasons
- [ ] Summer
- [ ] Winter

Countries
- [ ] Afghanistan
- [ ] Albania
- [ ] Algeria
- [ ] American Samoa
- [ ] Andorra
- [ ] Angola
- [ ] Antigua and Barbuda
- [ ] Argentina

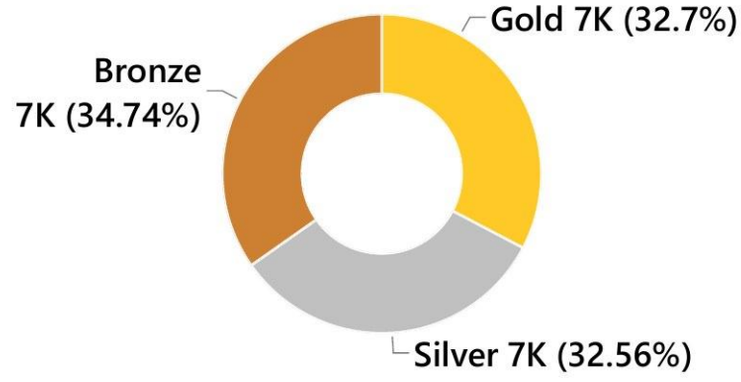
Year-wise Medal Count for Country



Best Performing Sports by Country



Medal Distribution



Global Medal Overview | Athlete Spotlight | Sport & Event Analysis | Host Country Analysis | Gender & Demographics

Country Insights: Insights

1 13.25 medals per 100 athletes

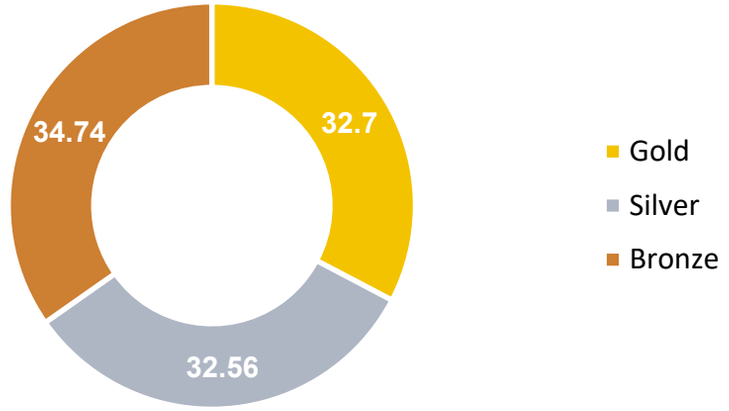
With 20.65K medals across 155.87K athletes, the average medal efficiency is 13.25. This is the yardstick to judge any single country against.

13.25 medals per 100 athletes

2 Medals are shared almost evenly by colour

Gold 32.7%, Silver 32.56%, Bronze 34.74%. Bronze edges ahead, which fits the extra bronzes given in sports with tied or double bronze matches.

Medal distribution (%)



3 Five sports hold the medal table

Athletics (4.51K), Swimming (3.28K), Rowing (3.07K), Artistic Gymnastics (2.28K) and Football (2.06K). The top three alone account for just over half of all medals.

WHY IT MATTERS Lets planners compare a country's medal output with its team size and sport mix.

Athlete Spotlight

Career longevity, top performers, and demographics of medal winners

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Total Athletes
155.86K

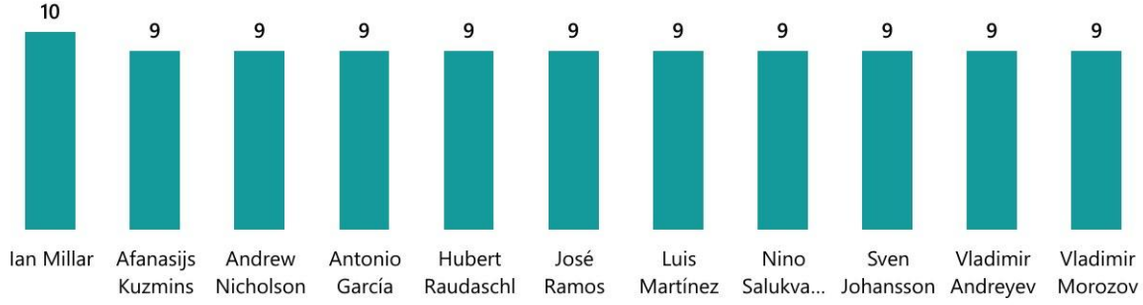
Medal Winning Athletes
32K

Medal Winning %
20.26%

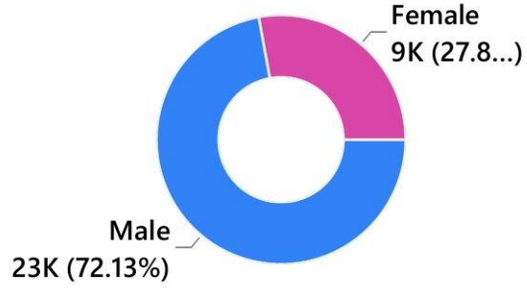
Medals Won by Athletes
44.69K

Timeline
1896 2032

Most Participated Athletes (Olympic Appearances)



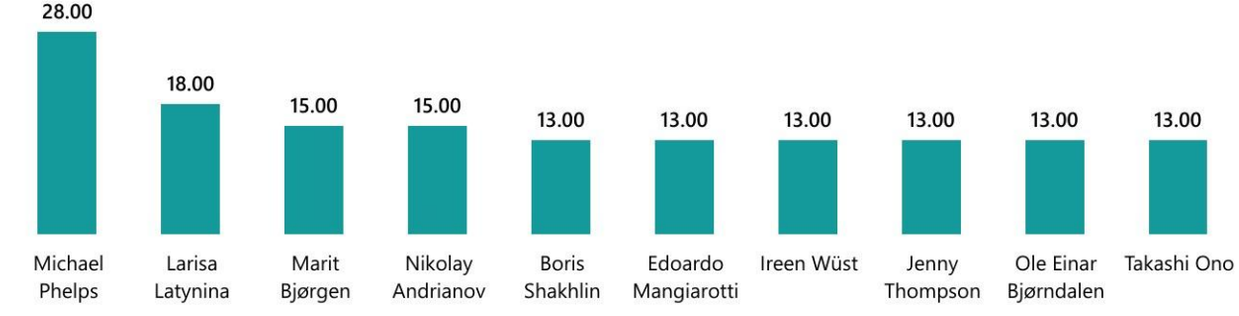
Gender Distribution of Medal Winners



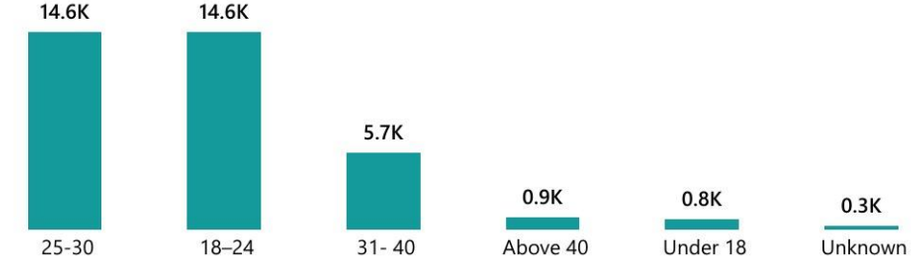
Gender
- Female
- Male

Seasons
- Summer
- Winter

Top 10 Athletes by Total Medals



Athlete Age Distribution at Medal Wins



Global Medal Overview | Country Insights | Sport & Event Analysis | Host Country Analysis | Gender & Demographics

Athlete Spotlight: Insights

1

One in five athletes reaches the podium

32K of 155.86K athletes won a medal (20.26%), together taking 44.69K medals, so winners average more than one medal each.

2

Legends are rare outliers

Michael Phelps leads with 28 medals, well ahead of Larisa Latynina (18) and Marit Bjørgen and Nikolay Andrianov (15 each). The top 10 stop at 13.

3

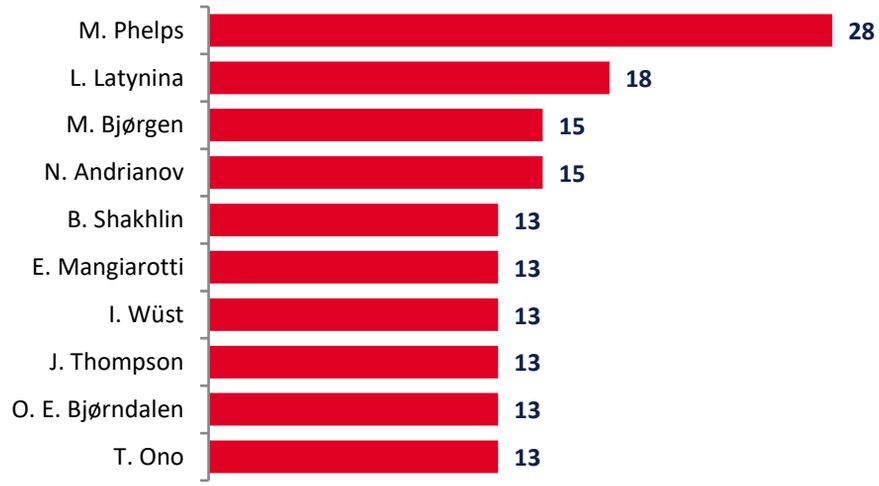
Peak age is 18 to 30

The 18–24 and 25–30 bands each hold 14.6K medal wins, about 79% of all age-tagged wins. Only 0.9K come after 40 and 0.8K before 18. Men are 72.13% of medal winners.

28

medals won by Michael Phelps, the all-time leader

Top 10 athletes by total medals



WHY IT MATTERS Shows what a successful career looks like: how long, how decorated and at what age.

Sport & Event Analysis

Where medals are concentrated — sport growth and country dominance

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Total Sports
112

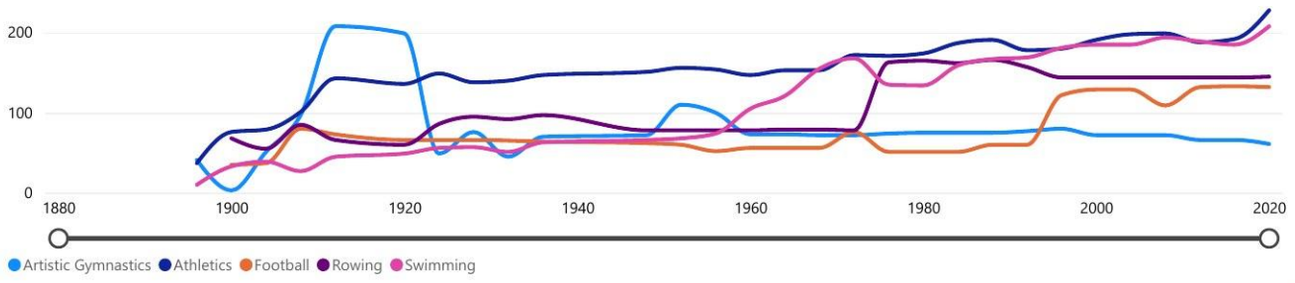
Total Events
964

Total Athlete Represented
155.87K

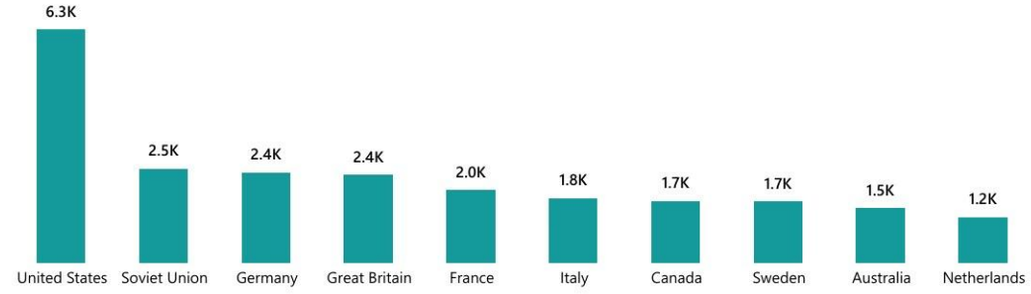
Seasons
- [ ] Summer
- [ ] Winter

Sports
- [ ] 3x3 Basketball
- [ ] Aeronautics

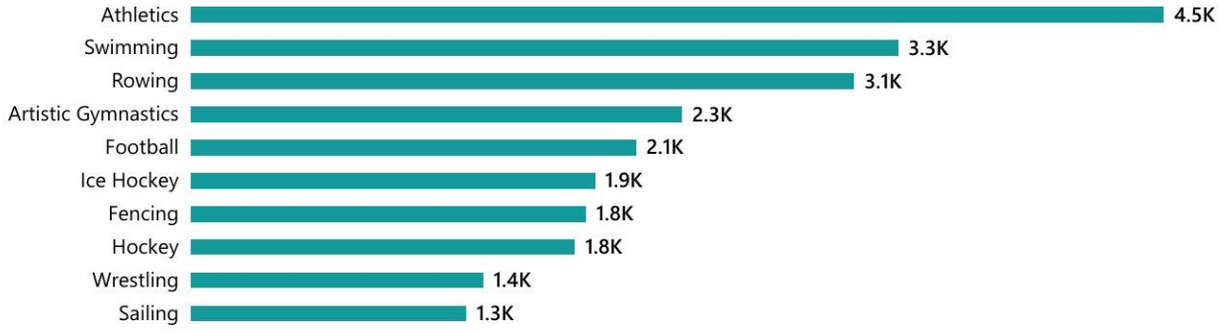
Sport-wise Medal Trends Over Time



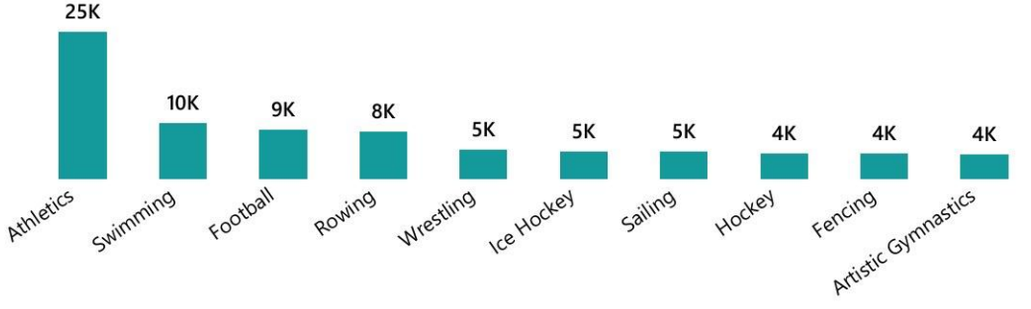
Country Dominance in Sports



Top Sports by Medals Awarded



Sport-wise Athlete Participation



Global Medal Overview | Country Insights | Athlete Spotlight | Host Country Analysis | Gender & Demographics

Sport & Event Analysis: Insights

1 Athletics is the Games' engine

25K athletes and 4.5K medals, more than double the next sport in participation (Swimming, 10K). Football (9K) and Rowing (8K) follow.

4.51K Athletics medals, the most of any sport

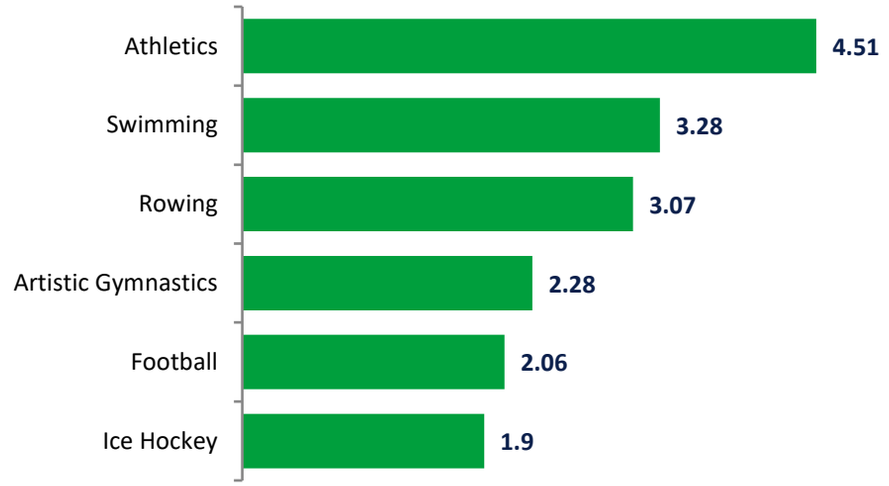
2 Some sports convert entries into medals far better

Rough medals-per-athlete ratios: Artistic Gymnastics about 0.57 (2.28K medals, 4K athletes), Rowing about 0.38, Swimming about 0.33, Football about 0.23, Athletics about 0.18.

3 Country dominance is concentrated

The United States has 6.3K sport-level medals, about 2.5x the Soviet Union (2.5K). Germany, Great Britain and France each sit near 2.0K–2.4K.

Top sports by medals awarded (K)



WHY IT MATTERS Guides where to invest: high-yield sports versus high-volume sports.

Host Country Analysis

Does hosting the Games come with a medal advantage?

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Olympics Hosted
57

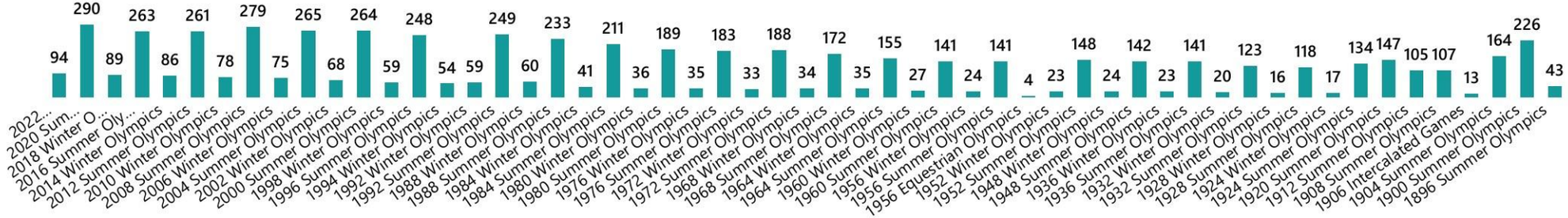
Total Medals
20.65K

Total Athlete Represented
155.87K

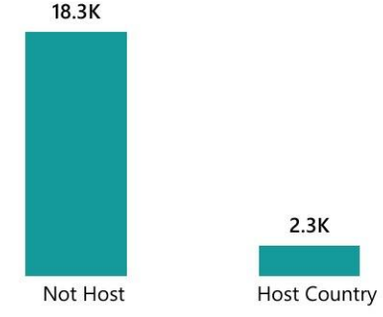
Seasons
- [ ] Summer
- [ ] Winter

Host Countries
- [ ] Australia
- [ ] Austria
- [ ] Belgium
- [ ] Brazil
- [ ] Canada
- [ ] Finland
- [ ] France
- [ ] Germany
- [ ] Great Britain

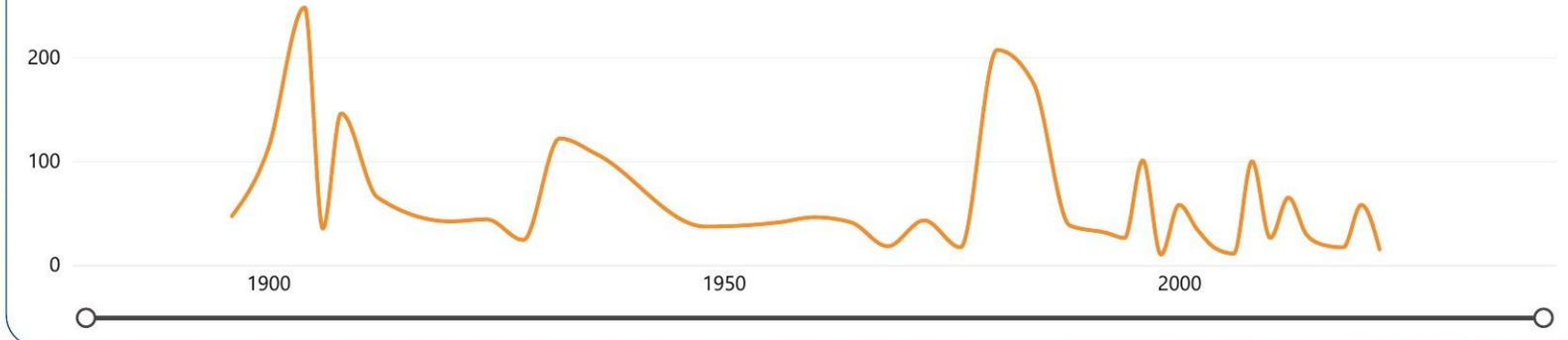
Events Conducted in Host Olympics



Medal Count as Host vs Non-Host



Host Country Medal Trend



Global Medal Overview | Country Insights | Athlete Spotlight | Sport & Event Analysis | Gender & Demographics

Host Country Analysis: Insights

1

Hosts win about 11% of all medals

Host editions account for 2.3K medals against 18.3K when not hosting, roughly 11% of the 20.65K total, from only a small share of country-editions.

2

Summer editions carry four times the events

Summer Games run about 150–290 events, Winter Games roughly 40–100. Larger programmes give hosts more chances to medal at home.

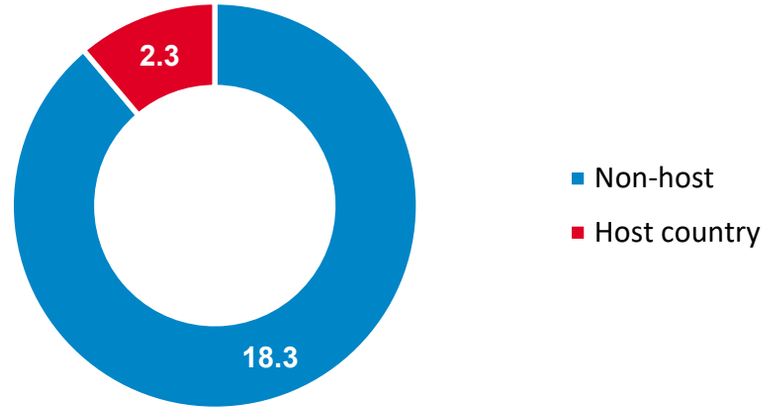
3

Home spikes are historic, not constant

Host-country medals peaked around 245 in the early 1900s and again around the 1980s, then settled below 100 in recent editions as competition broadened.

≈11% of all medals were won by host countries

Medals as host vs non-host (K)



WHY IT MATTERS Informs host bids: a home Games lifts medals, but the lift is shrinking.

Gender & Demographics

Tracking participation balance and athlete age over Olympic history

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Youngest Medal Winner
11

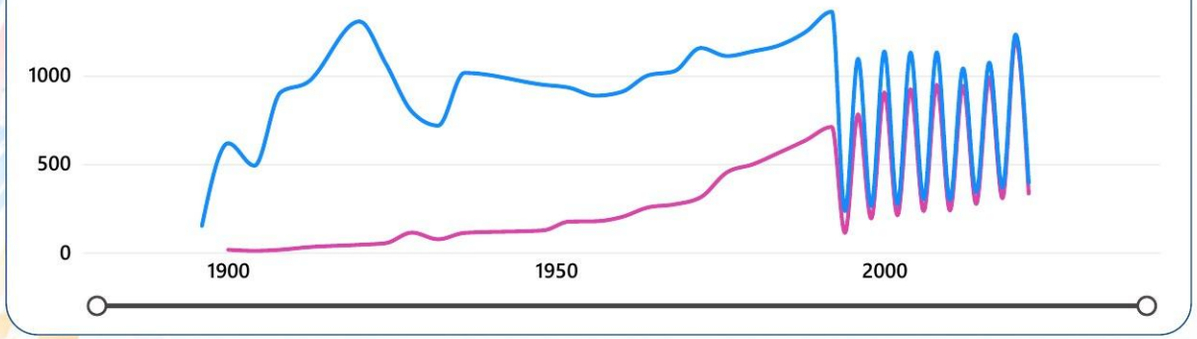
Oldest Medal Winner
73

Avg Age of Participants
26.05

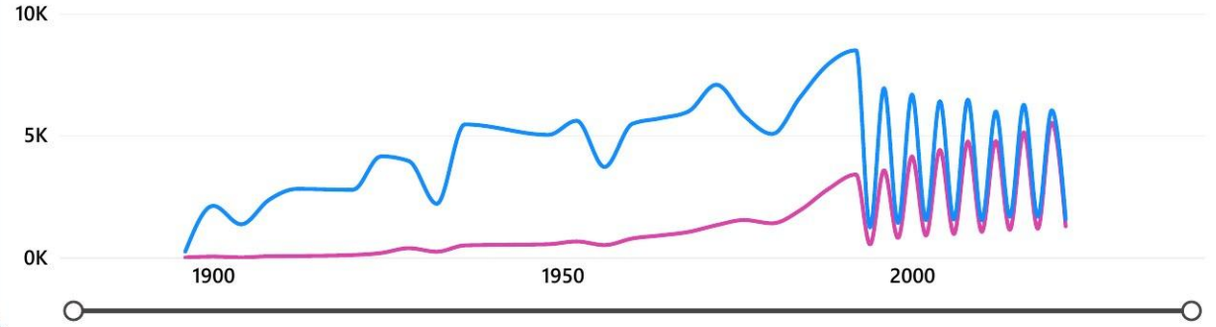
Timeline
1896 2032

Seasons
- [ ] Summer
- [ ] Winter

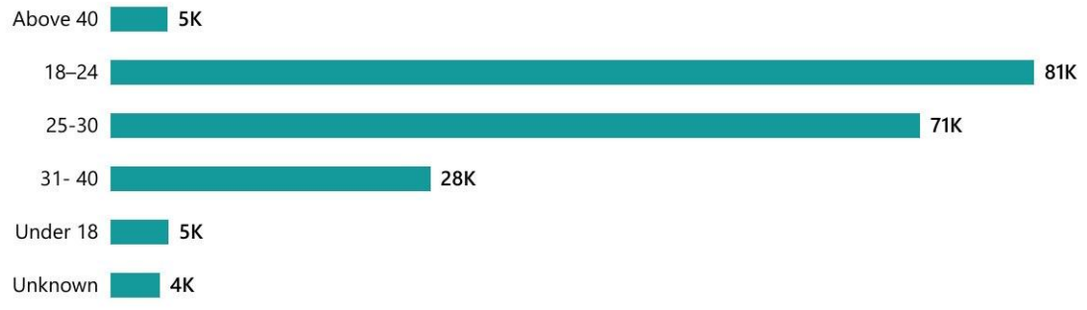
Gender-wise Medal Distribution Over Years



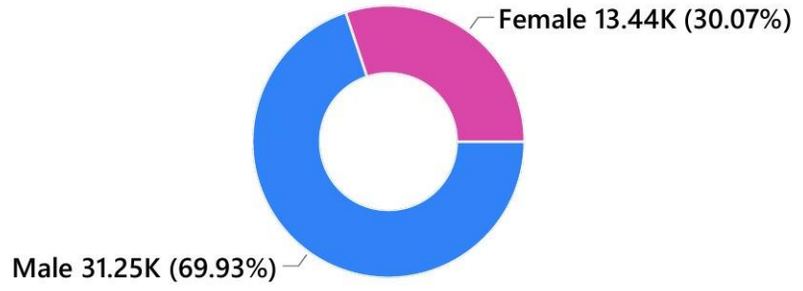
Male vs Female Athlete Count Over Years



Total Athlete Represented by age_band



Gender-wise Medal Distribution



Global Medal Overview | Country Insights | Athlete Spotlight | Sport & Event Analysis | Host Country Analysis

Gender & Demographics: Insights

1

Women now win about 30% of medals

Female athletes took 13.44K medals (30.07%) and males 31.25K (69.93%). The female line sits near zero before 1950 and climbs steadily, narrowing the gap.

30%

of all medals won by women (13.44K)

2

The Games belong to the young

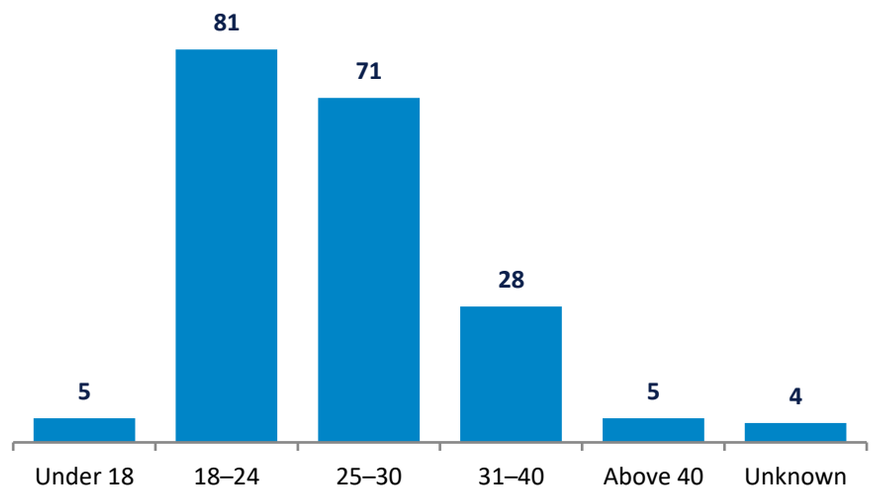
18–24 (81K) and 25–30 (71K) make up about 78% of athlete entries. The average participant is 26.05 years old.

3

Age extremes show the range

The youngest medal winner was 11 and the oldest 73. Fewer than 5K entries are under 18 and about 5K above 40.

Athletes by age band (K)



WHY IT MATTERS Tracks progress on gender balance and talent-pipeline age windows.

04

The Verdict

What the six dashboards say when read together, and what the IOC should do next.

SYNTHESIS

Key Findings Across the Six Dashboards

2.5x

Scale is concentrated

The US out-medals the Soviet Union 2.5 to 1; a handful of nations and sports hold most medals.

~1,000

Programme size shapes results

Summer editions reach ~1,000 medals and 290 events; Winter stays near 200–300 medals.

1 in 5

Podiums are rare, legends rarer

20.26% of athletes medal; the top individual has 28.

≈11%

Hosting helps, less than it used to

About 11% of medals come at home, with historic spikes fading.

18–30

Peak performance window

About four in five medal wins fall in this age range.

30%

The gender gap is closing

Women take 30% of medals, up from near zero before 1950.

WHAT THE IOC SHOULD DO

Final Recommendations

1 Back high-yield sports

Gymnastics, Rowing and Swimming convert athletes into medals best; track medals-per-athlete as a development KPI.

2 Broaden the medal base

Medals sit with a few nations. Support emerging NOCs in sports where efficiency is high.

3 Plan around the 18–30 window

Build pipelines that deliver peak athletes in their twenties, while supporting long careers.

4 Keep growing women's participation

Women win 30% of medals; targeted event additions can lift the share further.

5 Treat hosting as an investment

Home lift is real (~11% of medals) but shrinking; tie bids to sport-specific preparation.

6 Track KPIs every edition

Keep this report as the standing scorecard: efficiency, win rate, host share, gender split.

NEXT STEP Add forecasting (trend lines per country and sport) to deliver the "forecast future outcomes" part of the brief.

THANK YOU

Olympic Insights Using Data Analytics

Yatharth Aphale

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